

How AI Changes Red Teaming

A PRACTITIONER'S GUIDE FOR SECURITY LEADERS



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AI doesn't make the fundamentals less important. It makes ignoring them **more consequential**

Adversaries are using AI to move faster and reach further. The AI systems your organization is deploying have become targets with access built in by design. Most programs weren't built for either.

Today we'll discuss:

- What a red team engagement is designed to accomplish
- What it means when AI systems become targets
- How AI is changing adversary capabilities
- How SpecterOps' AI red team is operating: the research, lessons learned, and tools



What exactly is **AI red teaming**?

What red teaming should test

WHY IT MATTERS

- 'Red team' has become a catch-all, and 'AI red team' is even less defined. Without clarity on what's being tested, orgs can't evaluate what they're getting
- Pen testing reveals exposure; red teaming tests whether you'd detect and respond
- Done right, a red team engagement provides a chance to observe real adversary behavior and respond under realistic conditions

*What did your last "red team" **actually** test?*

Detection and response, or just whether something could be exploited?

*If your last engagement
started from the
perimeter, what did you
learn about **post-breach
activity?***

The case for assumed breach

WHY IT MATTERS

- Starting from the perimeter can consume the engagement without answering the most important questions
- Preventing all initial access is not a realistic objective
- The more consequential questions start after access is established
- AI-enabled adversaries move faster and reach further

Documentation isn't the same as capability

WHY IT MATTERS

- A playbook that has never been exercised is a hypothesis
- Detection and response is a performance discipline built through practice, not documentation
- Safe-to-fail exercises exist so gaps surface in practice, not during an actual incident

*Your team has a playbook.
Has anyone run it start to
finish under real
conditions, or is it **just a
hypothesis?***

*When did you last run an
exercise designed to **build**
capability, not just
measure it?*

Two different investments, two different outcomes

WHY IT MATTERS

- Engagements reveal gaps; exercises close them. They serve different purposes
- Most organizations invest regularly in measurement; structured practice gets deprioritized
- Capability gaps compound if the only feedback loop is assessment. And, most organizations haven't measured or practiced response for AI-specific attack scenarios

Findings only matter if they drive decisions

WHY IT MATTERS

- Vague goals produce findings that are technically valid but difficult to prioritize
- An objective is actionable when its answer connects to a decision the organization is prepared to make
- One clear primary objective keeps the engagement focused and produces findings leadership can act on

*When your last engagement ended, did the findings connect to a decision your organization was **ready to make**?*

Now apply all of that to an
environment where AI is both a
tool **and a target**

What's the **impact** if your AI system is compromised?

- AI agents are granted **broad** access by design: files, APIs, browser, emails, communication tools
- A compromised agent may have numerous lateral movement paths
- Most organizations have no baseline for what an AI system compromise looks like or how far it can reach

What's unique about testing **AI systems**?

- The fundamentals of red teaming hold but these characteristics are unique to AI:
 - Non-determinism
 - Tokenization
 - Context window
 - Memory

Questions to ask about AI systems in your environment

What **identities**
does it have?

What can it **do**?

Can we **detect**
abuse or
misbehavior?

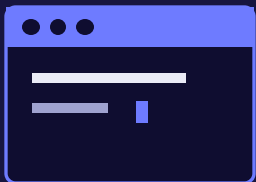
What can it
read?

What **impact**
could it have on
downstream
systems?

Will our
response work?

“Ways” of Testing

Behavior



MODEL-SPECIFIC TEST

WHAT IT TESTS

Model behavior during inference

KEY QUESTION

Can the model be induced to produce unsafe, unintended, or privacy-impacting output?

EXAMPLES

Alignment & safety · Prompt manipulation ·
Harmful output · Model leakage

EVIDENCE

Reproducible behavioral results

“Ways” of Testing

Behavior → Integrated Application



MODEL-SPECIFIC TEST

WHAT IT TESTS

Model behavior during inference

KEY QUESTION

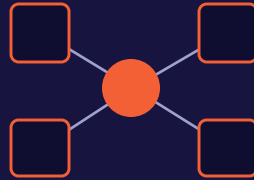
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AI SYSTEM SECURITY ASSESSMENT

WHAT IT TESTS

The integrated AI application & workflow

KEY QUESTION

What can an attacker make the system access or do?

EXAMPLES

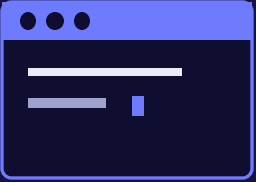
Authorization · RAG & data access · Tools & APIs · Agents · Memory · Downstream systems

EVIDENCE

Attack paths and demonstrated impact

“Ways” of Testing

Behavior → Integrated Application → Detection & Response



MODEL-SPECIFIC TEST

WHAT IT TESTS

Model behavior during inference

KEY QUESTION

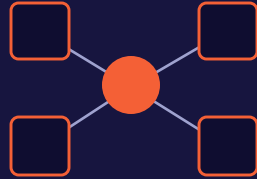
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AI SYSTEM SECURITY ASSESSMENT

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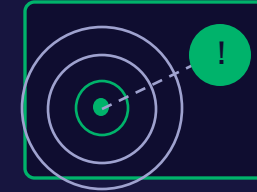
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EXAMPLES

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EVIDENCE

Attack paths and demonstrated impact



RED TEAM ENGAGEMENT

WHAT IT TESTS

Enterprise detection, response & resilience

KEY QUESTION

Would defenders detect and respond before an adversary achieved a meaningful objective using AI systems in your environment?

EXAMPLES

Assumed breach · Attack-path execution · Detection & response validation

EVIDENCE

Attack timeline, detection gaps, response actions & capability improvements

How does AI change the way **adversaries** operate?

AI makes **adversary operations** faster, broader, and more ephemeral

- AI-enabled adversaries systematically enumerate misconfigurations across every connected system at scale
- Accelerated vulnerability discovery, finding exploits defenders have no baseline for
- Ephemeral tool development, making signatures irrelevant
- Most programs were built to detect and respond to human-paced attacks

Where our AI red team is operating

PATTERNS

- Vulnerability and Exploit development is now common, and leads to new attack surfaces
- Going from idea to proof-of-concept is accelerated
- Defense evasion research has been reduced from weeks to hours

RESEARCH

- [Ghostworks: Cutting Edge AI Research Division](#)
- [Building an Indirect Prompt Injection Workflow](#)
- [Disposable Tooling: LLM-Generated Mythic Agents](#)
- [Accelerating EDR Evasion with LLM-Driven Analysis](#)
- [Blacklight: AI Agent Endpoint Artifacts](#)

TOOLS

- [Blacklight](#) — AI agent endpoint assessment
- [Sage](#) — Mythic C2 sub-agent coordination (research prototype)
- [Jailbreaker](#) — structured LLM jailbreak testing
- [AI-Gated Loader](#) — model-gated execution, fail-closed
- Ghostwriter + Mythic — AI-assisted attack narrative drafting, active across engagements

What our AI red team is seeing

Windows CVEs per Patch Tuesday

Total CVEs assigned per month · Sep 2023 – Sep 2026



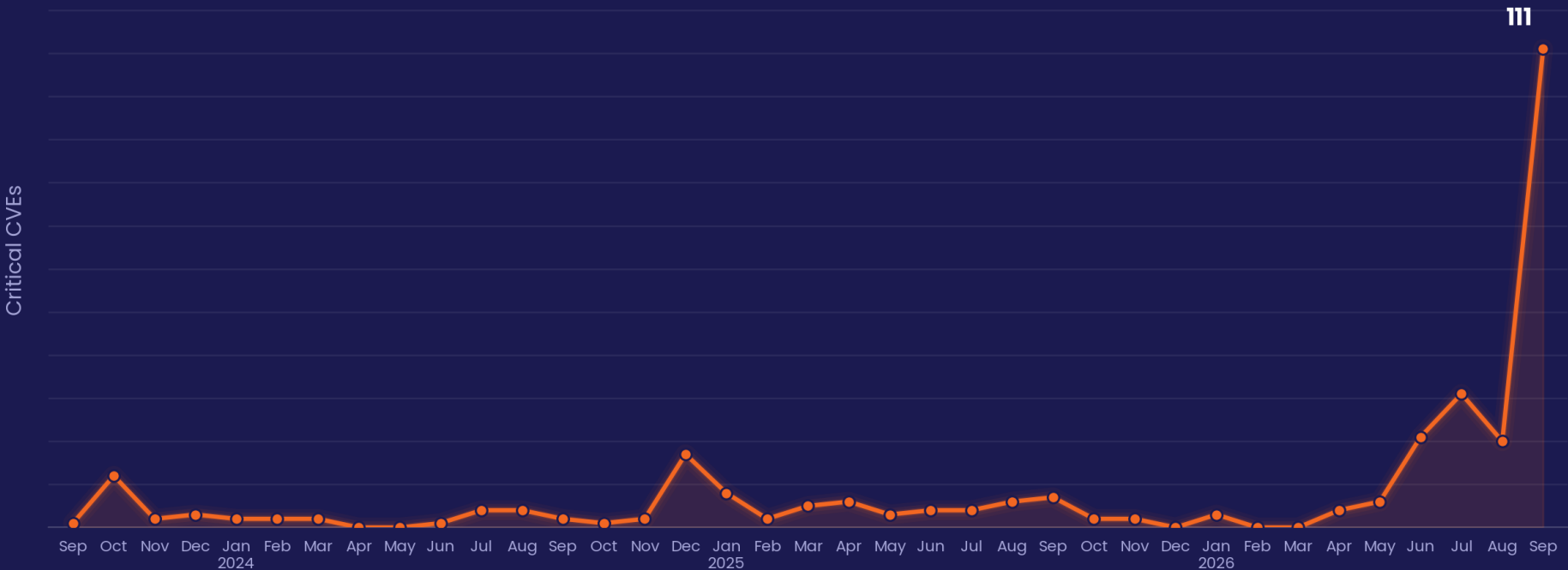
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data from <https://api.msrc.microsoft.com/cvrf/v3.0/cvrf/>

What our AI red team is seeing

Critical Windows CVEs per Patch Tuesday

CVEs rated Critical each month · Sep 2023 – Sep 2026



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Project Nucleon

Born out of the question:

What would a modern attack platform look like if it was built for AI from the ground up?

Ongoing research is being used to address:

- Disposable Tooling: Building LLM-Generated Mythic Agents from Prompt to Deployment
- Accelerating EDR Evasion with LLM-Driven Analysis

Current limitations of AI:

- Consistent feedback is required to steer LLM progress
- Hard to work with existing tooling which doesn't surface telemetry

Project Nucleon is an internal project which provides a framework designed for LLMs from the ground up

Summary

AI has not
changed the
fundamentals of
red teaming

AI systems are
yet another
adversary target

AI enables
adversaries to
operate at ever
increasing
speed and scale

Red teams are
adapting to
accurately
simulate today's
AI-enabled
adversary

Q&A

Please use the built-in Q&A feature on the lower right-hand side now to submit your questions.

Specter Bash 2026

October 5-8



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- Adversary Tactics: Red Team Operations
- Adversary Tactics: Identity-driven Offensive Tradecraft
- Adversary Tactics: Detection
- Adversary Perspectives: Azure



Adversary Intelligence: LLM Tradecraft

Build, evaluate, attack, and defend LLMs and agentic systems

Participants will learn:

- How to build and evaluate LLMs and agentic systems, including how to assess their architecture, limitations, and security implications in real enterprise environments
- How to identify and understand attacker techniques targeting LLMs and agentic systems, including prompt injection, jailbreaks, and exploitable weaknesses in AI infrastructure and agent design
- How to apply AI to defensive security workflows, including threat modeling, security testing, and AI-assisted reverse engineering using leading AI technologies

Sign up starting Sept 22!

